AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

Business Statistics Paper II

Time: 1 hour 30 minutes Marks: 30

AKEB May 2018
Test for E&L Learning Only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are SEVEN questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 4 Marks)

i. Define primary and secondary data. (2 Marks)

ii. Enlist any FOUR methods of collection of primary data. (2 Marks)

AKU-EB May 2018 for Teaching & Learning Only

Q.2. (Total 4 Marks)

The given frequency distribution represents the weight (in kilograms) of 50 employees at an organisation.

Weight (kg)	Number of Employees
40 – 49	3
50 – 59	11
60 – 69	17
70 – 79	14
80 – 89	5

Find the

i. class width of the given frequency distribution. (1 Mark)

ii. class boundary of the class interval with the highest frequency. (1 Mark)

iii. class mark of the class interval 50 – 59. (1 Mark)

iv. cumulative frequency of the class interval 50 – 59. (1 Mark)

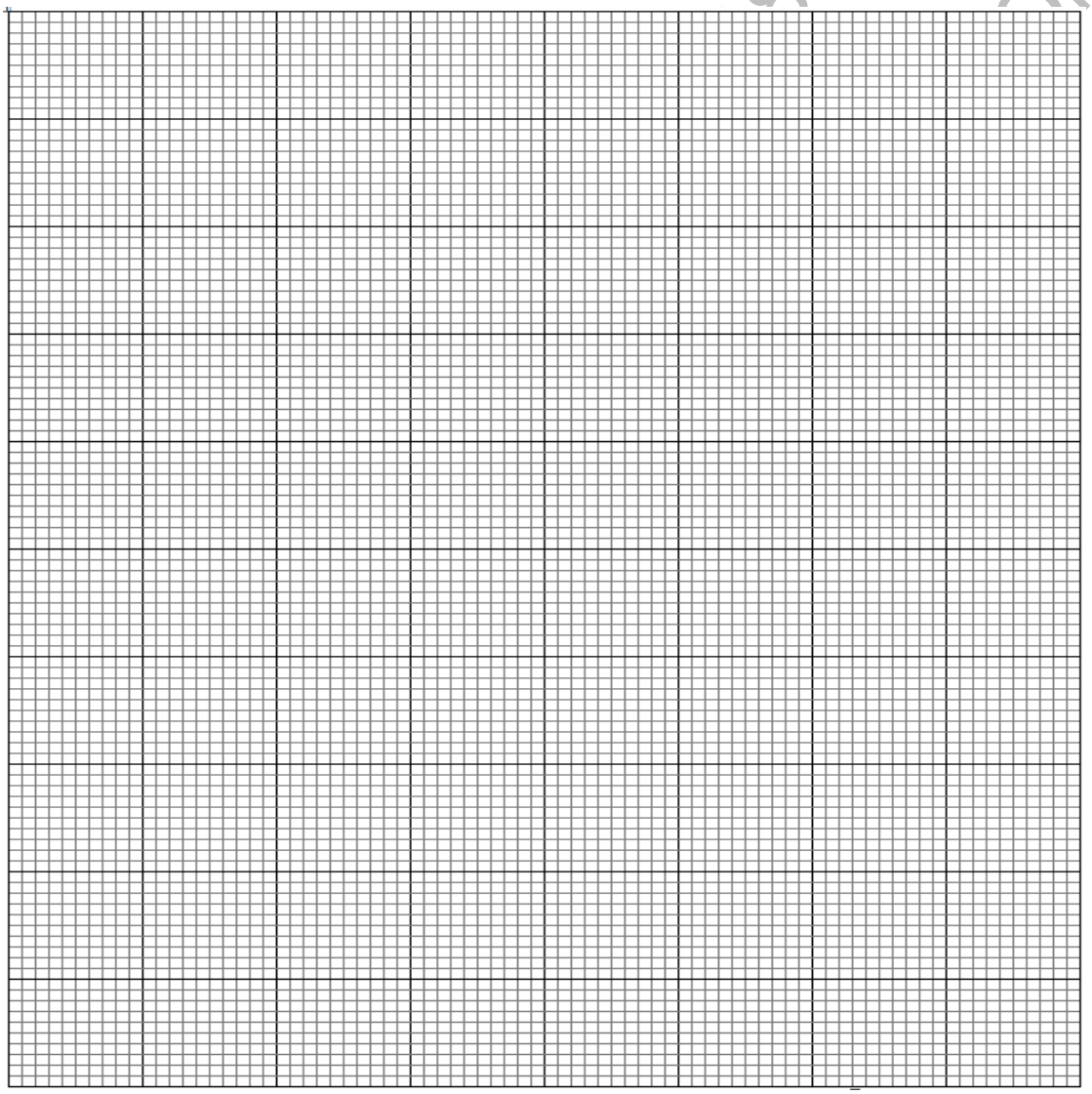
AKU-EB May 2018 for Teaching & Learning Only

Q.3. (Total 4 Marks)

The given frequency distribution represents the weight (in kilograms) of 100 students in a school.

Find the class boundaries of each class interval and represent the given data using a histogram.

Weight (kg)	Number of Students	Class Boundary
40 – 49	6	
50 – 59	22	
60 – 69	34	
70 – 79	28	
80 – 89	10	



(**ATTEMPT EITHER PART a OR PART b OF Q.4.)**

Q.4. (Total 5 Marks)

a. The weights (in kilograms) of 150 members of a health club, who are following a diet plan, are represented by the given frequency distribution.

i. Complete the given table to find the mean weight of the club members. (4 Marks)

Weight (kg)	**Number of Members**		
55– 64	22		
65– 74	38		
75 – 84	50		
85 – 94	23		
95 – 104	17		
Total		–	

AKU-EB May 2018 for Teaching & Learning Only

ii. If each member of the club reduces 2 kg weight, find the mean of the reduced weights using your answer of part i. (1 Mark)

(**ATTEMPT EITHER PART a OR PART b OF Q.4.)**

b. The following are the sales (in Rupees) of tomatoes in twenty days. (5 Marks)

310	325	110	175
290	110	150	280
230	170	100	195
180	180	175	145
230	250	175	180

i. Find the mean, median and mode of the given sales. (4 Marks)

ii. If each sale in the given data is doubled, then what will be its impact on the median of the data? (1 Mark)

AKU-EB May 2018 for Teaching & Learning Only

Q.5. (Total 4 Marks)

Samina is fond of reading books. She noted her reading time for the first 12 days of a month. The time noted (in hours) is as follows:

Time (hours)	2	3	6	8	6.5	4.5	3	3.5	2	4	5	6

Complete the table to find the variance and standard deviation of her reading time.

AKU-EB May 2018 for Teaching & Learning Only

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a.

i. Define the following terms.

I. Index number (1 Mark)

II. Simple index number (1 Mark)

III. Composite index number (1 Mark)

ii. Define base period in the construction of an index number. Also write any ONE of its characteristics. (2 Marks)

AKU-EB May 2018 for Teaching & Learning Only

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. In the following table, the prices and quantities of three commodities in the years 2008 and 2009 are given. Construct the base year weighted index by taking 2008 as base year. Also comment on the result. (5 Marks)

Commodities	Prices (Rs/ kg)		Quantities (kg)			
	2008	2009	2008	2009		
X	16	20	40	50		
Y	12	15	90	100		
Z	14	16	50	60		
Total						

AKU-EB May 2018 for Teaching & Learning Only

(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 4 Marks)

a. ALGORITHMS is a ten-letter word.

i. How many passwords of 6 letters can be formed from the letters of the word ALGORITHMS, if

I. each letter is used once? (1 Mark)

II. repetition of letters is allowed? (1 Mark)

ii. How many passwords of 10 letters can be formed from the letters of the word ALGORITHMS, if

I. each letter is used once? (1 Mark)

II. repetition of letters is allowed? (1 Mark)

AKU-EB May 2018 for Teaching & Learning Only

(ATTEMPT EITHER PART a OR PART b OF Q.7.)

b.

i. Write the sample space of an experiment when two coins are tossed and a die is rolled simultaneously. (2 Marks)

(**Note:** Plural of die is dice.)

ii Hence, find the probability of getting

I. two heads and an odd number. (1 Mark)

II. exactly one head and an even number. (1 Mark)

AKU-EB May 2018 for Teaching & Learning Only

END OF PAPER

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only